

Scale Immersive: 3D Model Input Requirements

This document defines the requirements that contributor `.blend` files must meet before they are submitted to the Scale Immersive processing pipeline. Following these requirements ensures predictable bake results, stable runtime performance, and visually correct output on the Apple Vision Pro.

Pipeline Overview

The Scale pipeline takes a fully lit Blender scene and bakes the complete lighting, shading, and material response into texture maps. The original lit materials are then replaced at runtime with a lightweight **unlit shader** that simply samples the baked textures. This baked, unlit output is what is rendered on the Apple Vision Pro.

Because all lighting is "frozen" into textures, the quality of the final result depends almost entirely on the quality of the input scene: good light setup, nice textures, clean geometry, correct normals, well-laid-out UVs, and consistent naming. Issues that might be invisible in a real-time viewport — bad normals, overlapping UVs, hidden interior geometry, degenerate faces — will become permanent artifacts once baked.

The requirements below exist to make the bake deterministic and the runtime experience clean.

1. Triangle Count

- **Maximum 5 million triangles per scene.**
- If a location naturally exceeds this budget, split it into multiple logical scenes (e.g. `Floor1`, `Floor2`, `Basement`, `Exterior`) rather than decimating beyond recognition.
- Splits should follow real spatial boundaries that make sense for navigation and streaming, not arbitrary mesh cuts.

2. Optimization

- Strip away all geometry the user will never see. This includes the **back sides of walls**, the **undersides of carpets and rugs**, the **backs and undersides of furniture against walls**, and any internal volumes of solid props.
- The scene must contain **no degenerate geometry**: no zero-area faces, loose vertices, loose edges, duplicate vertices, or interior faces.
- **No overlapping or duplicate surfaces.** Two coplanar faces occupying the same space cause z-fighting in the viewport and produce baking artifacts (flickering shadows, inconsistent AO, seam lines) that become permanent in the baked textures. Merge, delete, or offset any stacked geometry before submission.

- Meshes must be optimized for real-time rendering. Avoid excessive polygon density on objects that do not visually require it (a flat wall does not need to be subdivided, a smooth cylinder rarely needs more than 24–32 sides, etc.).
- Run a mesh cleanup pass before submission (e.g. Merge by Distance, Delete Loose, Select Interior Faces) and verify the result.

3. Normals

- All normals must be correct and consistently oriented (outward-facing for solid objects).
- The Vision Pro renders with **backface culling enabled**, which means any inward-facing or flipped face will appear as a hole in the final scene.
- The scene must be visually **watertight from every position the user can occupy**. Walk the camera through the intended user path and confirm there are no missing faces, flipped normals, or visible "inside-out" surfaces.
- **Normal map images must have their color space set to Non-Color** in the image properties. Normal maps left in **sRGB** will bake incorrectly.

4. UV Maps

Every object in the scene must have **two UV sets**:

1. **UVMap** (*active*) — used for tiling and general texturing. This is the standard authoring UV set.
2. **SimpleBake** — used as the bake target. This UV set must be:
 - Fully unwrapped
 - **Non-overlapping** (no two faces sharing the same UV space)
 - Laid out as **as few and as large** islands as possible, to maximize texel density
 - Within the 0–1 UV space
 - Packed with a small island margin (the pipeline uses **0.001** by default — leave at least this much spacing between islands to avoid bleed)

For large or visually important meshes — sofas, floors, walls, large furniture, kitchen countertops, etc. — add UV seams thoughtfully so the unwrap stays efficient and avoids stretched or fragmented islands. Texel density on these surfaces directly determines how sharp the baked result looks at viewing distance.

5. Structure & Naming

Naming Convention

- All objects must have human-readable names in **PascalCase**.
- Examples: **LampShadeGreen**, **LampShadeRed**, **CounterTop**, **Floor**, **WoodenFrameNorth**.
- No default names (**Cube.001**, **Mesh**, **Object**, etc.) anywhere in the scene.

- Names must only contain **alphanumeric characters, underscores, dots, or hyphens** (`a-z` , `A-Z` , `0-9` , `_` , `.` , `-`). No spaces, accented characters, slashes, parentheses, or other special characters — these will be stripped or replaced by the pipeline.
- For mesh objects, the underlying **mesh data block must share the same name as the object** (e.g. object `LampShadeGreen` uses mesh data `LampShadeGreen` , not `Mesh.042`).

One Material Per Object, Matching Names

- The scene must be split by material: every object has exactly one material, and the **object name and material name must match**.
- Example: an object named `LampShadeGreen` has a material named `LampShadeGreen` .
- No empty material slots.** Every material slot on every object must contain a material reference; remove any unused slots before submission.

No Duplicates

- No `.001` , `.002` , etc. suffixes on either object or material names.
- If two objects legitimately need to share a material, they should be joined or renamed and re-materialied so each unique material appears exactly once in the scene.

Reserved Naming Keywords

The pipeline scans object and material names for specific keywords (case-insensitive) and applies special handling. Use them when relevant; avoid them when not.

Category	Keywords (any of)	Pipeline behavior
Glass	<code>glass</code> , <code>window</code> , <code>windows</code> , <code>lens</code> , <code>lenses</code>	Hidden during AO baking; receives a transparent shader treatment at runtime.
Foliage	<code>leaf</code>	Gets foliage shader treatment (subtle translucency).
Stitching	<code>stitch</code> , <code>stitches</code> , <code>stitching</code> , <code>seam</code> , <code>thread</code>	Hidden during AO baking to prevent self-shadowing artifacts.
Blackout	<code>blackout</code> , <code>blackoff</code> , <code>black_out</code> , <code>black_off</code> , <code>blackscreen</code> , <code>black_screen</code>	Excluded from AO/lighting baking entirely; remains visible.
Metal	<code>metal</code>	Receives a metallic shader treatment at runtime (reflective response).

Examples:

- Glass: `WindowsGlass` , `VaseGlass` , `CoffeeTableGlass` , `LampLens`
- Foliage: `FlowerLeaf` , `TreeLeaf` , `PlantLeaf`

- Stitching: `SofaStitch` , `JacketSeam`
- Blackout: `BlackoutPanel` , `BlackScreen`
- Metal: `FaucetMetal` , `DoorHandleMetal` , `LampBaseMetal`

Important: these keywords are reserved and matched as case-insensitive substrings anywhere in the name. Do not use them in object names where the special behavior is not wanted — pick a name that avoids the keyword entirely (e.g. a wooden window frame should be called `WoodenFrame` , not `WindowFrame` , since any name containing `window` will trigger glass treatment).

No Instances or Live Modifiers

- No linked duplicates, collection instances, or other instancing.
- No unapplied modifiers — including `Mirror` , `Subdivision Surface` , `Array` , `Solidify` , `Boolean` , etc. **All modifiers must be applied** before submission, so the geometry the pipeline sees is the geometry that will be baked.

6. Materials & Shaders

- All textures the materials reference must exist (no broken / pink "missing texture" links).

7. Transforms

- All objects must have **applied transforms**: location at `(0, 0, 0)` , rotation at `(0, 0, 0)` , scale at `(1, 1, 1)` .
- Use Blender's **Apply → All Transforms** on every mesh object before submission. Unapplied scale in particular causes incorrect normals, broken modifiers, and inconsistent texel density.
- Scene units should be **meters** (Blender default), at real-world scale. A doorway should be ~2 m tall, a coffee cup ~10 cm, etc.

8. Textures

- All textures must be **packed into the .blend file** (`File → External Data → Pack Resources`). Do not deliver loose texture files alongside the scene.

9. Mesh Quality

- Meshes should be **all-quads** wherever possible. Triangles are acceptable only where unavoidable; n-gons should be eliminated.
- Every mesh must be inspected in **MatCap view mode** before submission. MatCap reveals normal and surface irregularities (pinching, bad shading, hidden creases, non-planar faces) that are invisible in solid or material preview.

- Smoothing groups / shade smooth + auto smooth angles should produce clean, intentional shading — no faceted patches on surfaces meant to be smooth, no smoothed edges where a hard edge is intended.
-

Pre-Submission Checklist

Before handing off a `.blend` file to the Scale pipeline, confirm:

Geometry

- ☐ Triangle count is under 5M (or the scene is split logically)
- ☐ Hidden / invisible geometry has been removed
- ☐ No degenerate faces, loose verts/edges, or interior faces
- ☐ No overlapping / duplicate surfaces (z-fighting sources)
- ☐ All normals point outward; scene is watertight from user viewpoints
- ☐ All quads where possible; MatCap inspection passed

UVs

- ☐ Every object has both `UVMap` and `SimpleBake` UV sets
- ☐ `SimpleBake` is non-overlapping, packed, and texel-efficient

Naming & Structure

- ☐ All names are PascalCase, no `.001` suffixes, no special characters
- ☐ Object name == Mesh data name == Material name
- ☐ One material per object; no empty material slots
- ☐ Reserved keywords (Glass, Leaf, Stitch, Blackout, Metal) used only when intended

Materials & Textures

- ☐ All textures resolve (no missing / pink textures)
- ☐ All textures packed into the `.blend` file

Scene State

- ☐ All transforms applied (location, rotation, scale)
- ☐ All modifiers applied; no instances
- ☐ Scene is in meters at real-world scale